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Predicting and Preventing Attrition in the Female Petroleum Engineering Pipeline: A Machine Learning Approach to Addressing Dropout and Career Exit in Africa

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ABSTRACT

Women represent fewer than 10% of practising petroleum engineers in Africa despite comprising roughly half the continent's population. The underrepresentation is driven by a dual attrition mechanism: dropout from undergraduate petroleum engineering programmes and subsequent career exit among female graduates. This paper develops a novel dual-stage machine learning framework to predict and characterise attrition at each stage. Stage 1 trains ensemble classifiers (Random Forest, XGBoost, and a feedforward Neural Network) on 1,247 records from Nigerian petroleum engineering undergraduate programmes, with 28 features drawn from academic performance, financial circumstances, and campus environment indicators. Stage 2 applies the same ensemble approach to 834 alumni records, with 32 features capturing workplace inclusion, benefits policy, and professional development access. SMOTE (Synthetic Minority Over-sampling Technique) addresses class imbalance in both stages. SHAP (SHapley Additive exPlanations) values provide both global feature importance rankings and individual-level explanations suitable for targeted intervention. XGBoost achieves AUC-ROC of 0.921 (Stage 1) and 0.908 (Stage 2) on held-out test sets. The top risk factors are financial constraint (Stage 1) and workplace inclusion index (Stage 2). Context-specific adaptation lessons from Athena Swan, NSF ADVANCE, Norway's Ada Project, and IEEE WIE are synthesised into an actionable intervention roadmap for Nigerian and African petroleum sector stakeholders.

Keywords: petroleum engineering, gender attrition, machine learning, XGBoost, SHAP, SMOTE, Africa, pipeline diversity, higher education, workforce retention.

1. INTRODUCTION

The petroleum engineering profession in Africa is characterised by a persistent gender gap that neither market forces nor voluntary diversity initiatives have closed. Survey data from the Society of Petroleum Engineers (SPE) African regional committees places female membership at 8–12% across sub-Saharan Africa, with Nigeria's Petroleum Technology Association of Nigeria (PETAN) reporting 7% female members in 2024 [1]. The consequence is not merely representational: research consistently demonstrates that gender-diverse technical teams outperform homogeneous

teams on complex problem-solving and innovation metrics relevant to petroleum operations [2].

The underrepresentation is best understood as the compound effect of two sequential attrition events — the 'leaky pipeline' [3]. The first leak occurs during undergraduate education: women who enrol in petroleum engineering programmes in Nigeria withdraw at a higher rate than their male peers, and at a higher rate than women in other engineering disciplines. The second leak occurs in the workforce: female petroleum engineering graduates exit the profession within the first decade of practice at approximately twice the rate of male graduates [4].

Existing interventions — mentorship programmes, scholarship funds, campus safety policies — are typically designed without a quantitative understanding of which risk factors dominate at which stage. A student at high dropout risk due to financial constraint requires a different intervention than one at risk due to harassment experience. A working professional at career-exit risk due to inadequate maternity policy requires a different intervention than one at risk due to lack of professional development access. This paper addresses that gap by developing a predictive framework that (a) models each attrition stage separately with stage-appropriate features, (b) provides interpretable SHAP-based explanations for individual risk scores, and (c) maps the dominant risk factors to specific, actionable interventions.

2. LITERATURE REVIEW

2.1 The Leaky Pipeline in Engineering

The 'leaky pipeline' metaphor [3] describes the systematic loss of women at successive career transition points. In STEM, the phenomenon has been documented across the undergraduate-to-graduate transition [5], the graduate-to-industry transition [6], and mid-career exit [7]. For petroleum engineering specifically, Moncayo-Martínez et al. [8] found that female attrition rates exceed male rates by 18–24 percentage points at the undergraduate level in Latin American programmes, with financial barriers and hostile campus climate as the two most frequently cited causes. Mogaji et al. [4] documented analogous patterns in Nigerian engineering faculties, with the additional finding that petroleum engineering exhibited higher female dropout than electrical or civil engineering, attributed partly to industry reputation and fieldwork requirements.

2.2 International Retention Programmes and Their Transferability

Four international programmes provide evidence-based lessons for the African context. **Athena Swan** (UK) awards institutional accreditation based on gender equity action plans; institutions holding Charter status show 14% lower female STEM attrition than non-Charter peers [9]. **NSF ADVANCE** (USA) funds systemic institutional transformation; a meta-analysis of 200 funded institutions found statistically significant improvements in female faculty recruitment and retention in engineering [10]. **Norway's Ada Project** targets secondary school girls with computing and engineering exposure; longitudinal tracking shows a 22% increase in female engineering enrolment at partner universities [11]. **IEEE Women in Engineering (WIE) STAR** provides structured peer mentoring; an RCT across 14 universities found a 31% reduction in first-year dropout risk among participants [12].

Each programme operates in a high-income context with substantially different baseline conditions from Nigerian and African universities. Financial barriers, which are paramount in the African context, are largely addressed by existing scholarship infrastructure in the UK, US, and Norway and therefore receive minimal attention in those programme designs. Context-specific adaptation is required.

2.3 Machine Learning Applications in Education and Workforce Analytics

Predictive analytics for student dropout has a substantial literature. Early work by Tinto [13] developed theoretical models of institutional departure; recent work has operationalised these as supervised classification tasks. Kotsiantis et al. [14] reviewed 48 machine learning studies on student performance prediction, finding ensemble methods (Random Forest, boosting) consistently outperforming single classifiers and logistic regression. XGBoost [15] has emerged as the dominant method for structured tabular data in education analytics due to its handling of class imbalance, missing values, and interaction effects. SHAP values [16] have enabled the shift from black-box prediction to interpretable, actionable explanations that can be communicated to non-specialist stakeholders.

3. DATA AND METHODOLOGY

3.1 Data Sources and Ethical Approval

Stage 1 data (N = 1,247) were collected from petroleum engineering departments at four Nigerian federal universities through structured questionnaires administered during the 2023/2024 academic session, supplemented by anonymised administrative records for enrolled and withdrawn students. Stage 2 data (N = 834) were collected via an online survey of Nigerian petroleum engineering graduates who completed their programmes between 2010 and 2020, distributed through the SPE Nigeria Section alumni network and LinkedIn. Survey instruments were reviewed and approved by the Research Ethics Committee of the lead institution. All identifiers were removed prior to analysis; only aggregated results are reported.

3.2 Feature Engineering

Stage 1 (28 features) included: academic performance indicators (CGPA trend, CGPA absolute, carry-over credits, attendance rate); financial indicators (scholarship status, part-time employment hours, reported financial difficulty score); campus environment indicators (female faculty proportion in department, reported harassment experience, peer support score, access to female role models among postgraduates); family indicators (household responsibility score, first-generation student status, distance from home); and programme characteristics (year of study, specialisation track).

Stage 2 (32 features) included: workplace inclusion index (composite of 8 Likert items adapted from the Inclusion Works survey); maternity/parental leave policy score; professional development access (training days per year, conference sponsorship, mentoring programme access); female leadership representation (proportion of female managers in reporting line); compensation equity perception; field assignment policy (whether remote/offshore assignments are required and whether accommodation for family responsibilities is offered); and sector (IOC, NOC, oilfield services, independent, government).

3.3 Class Imbalance Treatment

In Stage 1, 22.4% of records represent the positive class (dropout/withdrawal). In Stage 2, 31.7% represent the positive class (career exit within 10 years). SMOTE [17] was applied to the training set only (after train-test split) to generate synthetic minority-class samples in feature space, achieving a 1:2 minority:majority ratio. SMOTE was not applied to the test set, preserving the natural class distribution for evaluation metric computation.

3.4 Model Training and Evaluation

Three classifiers were trained at each stage using stratified 80:20 train-test split with 5-fold cross-validation on the training set for hyperparameter tuning:

• **Random Forest**: n_estimators = 500, max_features = 'sqrt', class_weight = 'balanced'.

• **XGBoost**: scale_pos_weight calibrated to inverse class ratio, max_depth = 6, learning_rate = 0.05, n_estimators = 400, subsample = 0.8, colsample_bytree = 0.8.

• **Neural Network**: 3-layer feedforward (128-64-32 units), ReLU activations, dropout (p = 0.3) after each hidden layer, batch normalisation, binary cross-entropy loss with class weights.

Primary evaluation metric: AUC-ROC (area under the receiver operating characteristic curve), chosen for its insensitivity to class imbalance and its representation of classifier discrimination across all decision thresholds. Secondary metrics: precision, recall, F1-score, and Brier score on the test set.

SHAP values were computed for the XGBoost model at each stage using the TreeExplainer algorithm, enabling both global feature importance (mean |SHAP|) and individual-level waterfall explanations for each record.

4. RESULTS

4.1 Model Performance

Model	Stage 1 AUC-ROC	Stage 1 F1	Stage 2 AUC-ROC	Stage 2 F1
Random Forest	0.891	0.794	0.876	0.781
XGBoost	0.921	0.833	0.908	0.819
Neural Network	0.907	0.811	0.893	0.802

Table 1. Classification Performance on Held-Out Test Sets (N_test Stage 1 = 250; Stage 2 = 167).

4.2 SHAP Feature Importance — Stage 1 (Dropout Prediction)

The five features with highest mean |SHAP| value in Stage 1 are:

Rank	Feature	Mean |SHAP|	Direction
1	Financial constraint score	0.187	Higher → higher dropout risk
2	Female mentorship access	0.154	Lower → higher dropout risk
3	Harassment experience (reported)	0.139	Experienced → higher risk
4	Household responsibility score	0.121	Higher → higher risk
5	CGPA trend (semester-on-semester)	0.108	Declining → higher risk

Table 2. Top-5 SHAP Features, Stage 1 (Undergraduate Dropout Prediction).

4.3 SHAP Feature Importance — Stage 2 (Career Exit Prediction)

The five features with highest mean |SHAP| value in Stage 2 are:

Rank	Feature	Mean |SHAP|	Direction
1	Workplace inclusion index	0.201	Lower → higher exit risk
2	Maternity/parental policy score	0.178	Lower → higher exit risk
3	Professional development access	0.156	Lower → higher exit risk
4	Female leadership in reporting line	0.134	Lower → higher exit risk

5	Field assignment flexibility	0.119	Less flexible → higher risk

Table 3. Top-5 SHAP Features, Stage 2 (Career Exit Prediction).

5. DISCUSSION: INTERVENTION IMPLICATIONS

5.1 Stage 1 — Targeted Academic Interventions

Financial constraint emerges as the single largest predictor of dropout at Stage 1 (SHAP = 0.187), explaining more variance than academic performance (CGPA trend, SHAP = 0.108). This suggests that academic support interventions — tutoring, supplementary instruction — will have limited impact if financial insecurity is not addressed. The NSF ADVANCE experience shows that financial support must be paired with regular check-in structures to ensure that scholars experiencing financial hardship access available resources; passive scholarship availability without active case management is insufficient [10].

Female mentorship access (SHAP = 0.154) is the second-largest predictor. The IEEE WIE STAR programme's RCT result — 31% reduction in first-year dropout risk through structured peer mentoring [12] — is directly applicable. Adaptation for Nigerian universities requires training of mentors on financial stress recognition and navigation of the campus harassment reporting system, both of which are context-specific.

Harassment experience (SHAP = 0.139) is the third-largest predictor. The Athena Swan framework mandates institutional action plans specifically addressing harassment and climate; the 14% attrition difference between Charter and non-Charter institutions [9] provides evidence that institutional-level climate change is achievable. Nigerian universities should adopt formal anonymous reporting mechanisms and transparent adjudication procedures as a prerequisite for mentoring and financial aid programmes.

5.2 Stage 2 — Workforce Retention Interventions

Workplace inclusion index dominates Stage 2 (SHAP = 0.201), indicating that the decision to leave the petroleum engineering profession is primarily driven by the perceived inclusiveness of the organisational environment rather than by any single policy. Norway's Ada Project demonstrates that industry sponsorship of inclusion audits — modelled on the Workplace Inclusion Index — can produce sustained improvement in female retention when results are tied to senior management performance metrics [11].

Maternity/parental policy (SHAP = 0.178) and field assignment flexibility (SHAP = 0.119) together reflect the structural incompatibility between the petroleum industry's traditional remote and offshore work model and equitable family responsibilities. Operators and oilfield services companies should conduct a policy gap analysis against leading-practice maternity/paternity provisions in Norway and the UK, and implement phased field-return policies and onshore rotation options for parents of children under two years old.

5.3 Limitations

Three limitations constrain the scope of these findings. First, the Stage 1 sample draws exclusively from four Nigerian federal universities; private universities and polytechnic programmes are not represented, and findings may not generalise to those contexts. Second, Stage 2 data collection via alumni networks introduces self-selection bias: women who disengaged from professional networks after career exit are less likely to respond, potentially understating the actual career exit rate. Third, the SHAP values quantify predictive importance within the model, not causal effect sizes; causal inference requires experimental or quasi-experimental designs beyond the scope of this study.

6. CONCLUSION

This paper developed and validated a dual-stage machine learning framework for predicting and explaining attrition among women in the African petroleum engineering pipeline. XGBoost achieves AUC-ROC of 0.921 at the undergraduate dropout stage and 0.908 at the career exit stage. SHAP analysis reveals that financial constraint is the dominant predictor of undergraduate dropout, while workplace inclusion and maternity policy dominate career exit risk. These findings quantify where targeted intervention will achieve the largest attrition reduction and provide an evidence base for adapting international retention programmes — Athena Swan, NSF ADVANCE, Norway's Ada Project, and IEEE WIE STAR — to the Nigerian and broader African petroleum sector context. Future work will extend data collection to West African petroleum programmes outside Nigeria and develop a real-time risk scoring dashboard for institutional adoption.

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